

QuantumSafe Sentinel - Executive Report

Scan ID: scan_e419cbbe

Executive Risk Score

Grade: F (0/100)

Findings Summary: 8 Critical | 6 High | 0 Medium | 0 Low

Migration Impact Estimate

Estimated Storage Bloat: 27.4048 MB/day

Handshake Latency Penalty: +15.0 ms per connection

Bandwidth Increase: 400.9%

Assumes 1000 ops/day. Sourced directly from local hardware benchmarks.

Vulnerability Detail & PQC Recommendations

Source	Current Algorithm	Context / Usage	Risk	PQC Recommendation
http01_example.py	RSA (Unknown)	code / misc	High	ML-KEM or ML-DSA (Usage Unclear - Verify)
http01_example.py	RSA (Unknown)	code / misc	High	ML-KEM or ML-DSA (Usage Unclear - Verify)
client_test.py	ECC (Unknown)	code / key_exchange	Critical	ML-KEM (Kyber) (Hybrid: X25519+Kyber768)
crypto_util_test.py	RSA (2048)	code / misc	High	ML-KEM or ML-DSA (Usage Unclear - Verify)

crypto_util_test.py	RSA (2048)	code / misc	High	ML-KEM or ML-DSA (Usage Unclear - Verify)
crypto_util.py	RSA (Unknown)	code / misc	High	ML-KEM or ML-DSA (Usage Unclear - Verify)
crypto_util.py	ECC (Unknown)	code / key_exchange	Critical	ML-KEM (Kyber) (Hybrid: X25519+Kyber768)
renewal_test.py	ECC (Unknown)	code / key_exchange	Critical	ML-KEM (Kyber) (Hybrid: X25519+Kyber768)
san_test.py	ECC (Unknown)	code / key_exchange	Critical	ML-KEM (Kyber) (Hybrid: X25519+Kyber768)
san_test.py	ECC (Unknown)	code / key_exchange	Critical	ML-KEM (Kyber) (Hybrid: X25519+Kyber768)
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san_test.py	ECC (Unknown)	code / key_exchange	Critical	ML-KEM (Kyber) (Hybrid: X25519+Kyber768)
misc.py	RSA (2048)	code / misc	High	ML-KEM or ML-DSA (Usage Unclear - Verify)
misc.py	ECC (Unknown)	code / key_exchange	Critical	ML-KEM (Kyber) (Hybrid: X25519+Kyber768)

Regulatory Exposure

The following frameworks and mandates are actively violated by findings in this scan:

CISA PQC Roadmap

Requirement: CISA directive urging agencies and critical infrastructure to catalog and prepare to replace vulnerable crypto.

Timeframe/Deadline: Must be fully cataloged immediately, transition before CRQC arrival.

Source: <https://www.cisa.gov/post-quantum-cryptography>

Violating Assets (14 findings):

http01_example.py - **RSA** (Risk: High)

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client_test.py - **ECC** (Risk: Critical)

crypto_util_test.py - **RSA** (Risk: High)

crypto_util_test.py - **RSA** (Risk: High)

crypto_util.py - **RSA** (Risk: High)

crypto_util.py - **ECC** (Risk: Critical)

renewal_test.py - **ECC** (Risk: Critical)

san_test.py - **ECC** (Risk: Critical)

san_test.py - **ECC** (Risk: Critical)

san_test.py - **ECC** (Risk: Critical)

san_test.py - **ECC** (Risk: Critical)

misc.py - **RSA** (Risk: High)

misc.py - **ECC** (Risk: Critical)

CNSA 2.0

Requirement: NSA's mandate for National Security Systems to transition to quantum-resistant algorithms.

Timeframe/Deadline: 2025 for software/firmware signing, 2033 full transition.

Source: https://media.defense.gov/2022/Sep/07/2003071834/-1/-1/0/CSA_CNSA_2.0_ALGORITHMS_.PDF

Violating Assets (8 findings):

client_test.py - **ECC** (Risk: Critical)

crypto_util.py - **ECC** (Risk: Critical)

renewal_test.py - **ECC** (Risk: Critical)

san_test.py - **ECC** (Risk: Critical)

san_test.py - **ECC** (Risk: Critical)

san_test.py - **ECC** (Risk: Critical)

san_test.py - **ECC** (Risk: Critical)

misc.py - **ECC** (Risk: Critical)